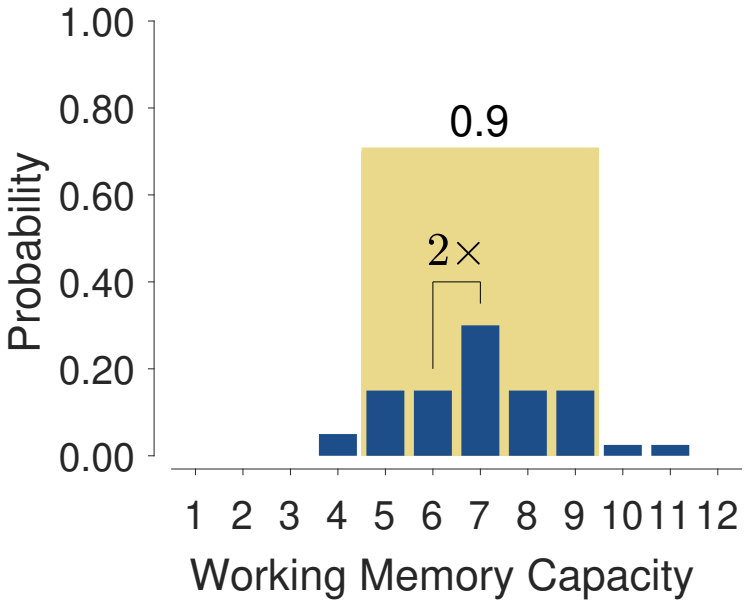
Bayesian Methods

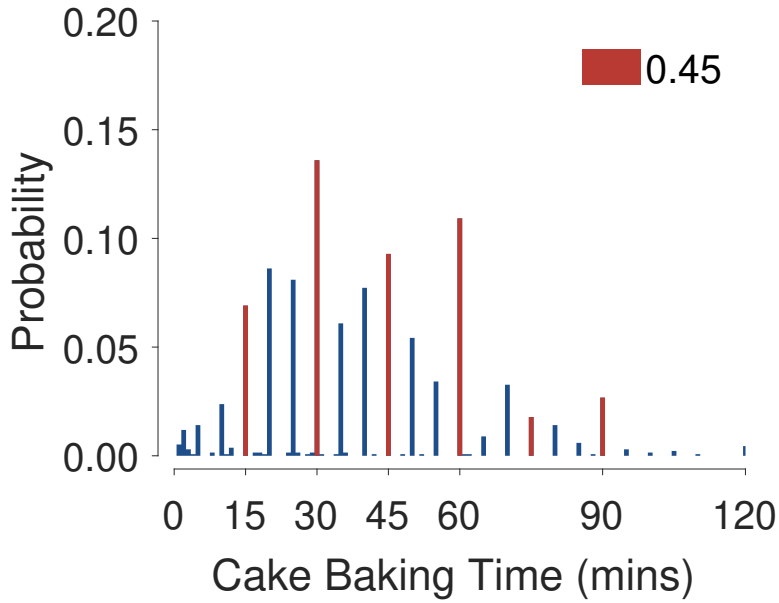
Michael D. Lee

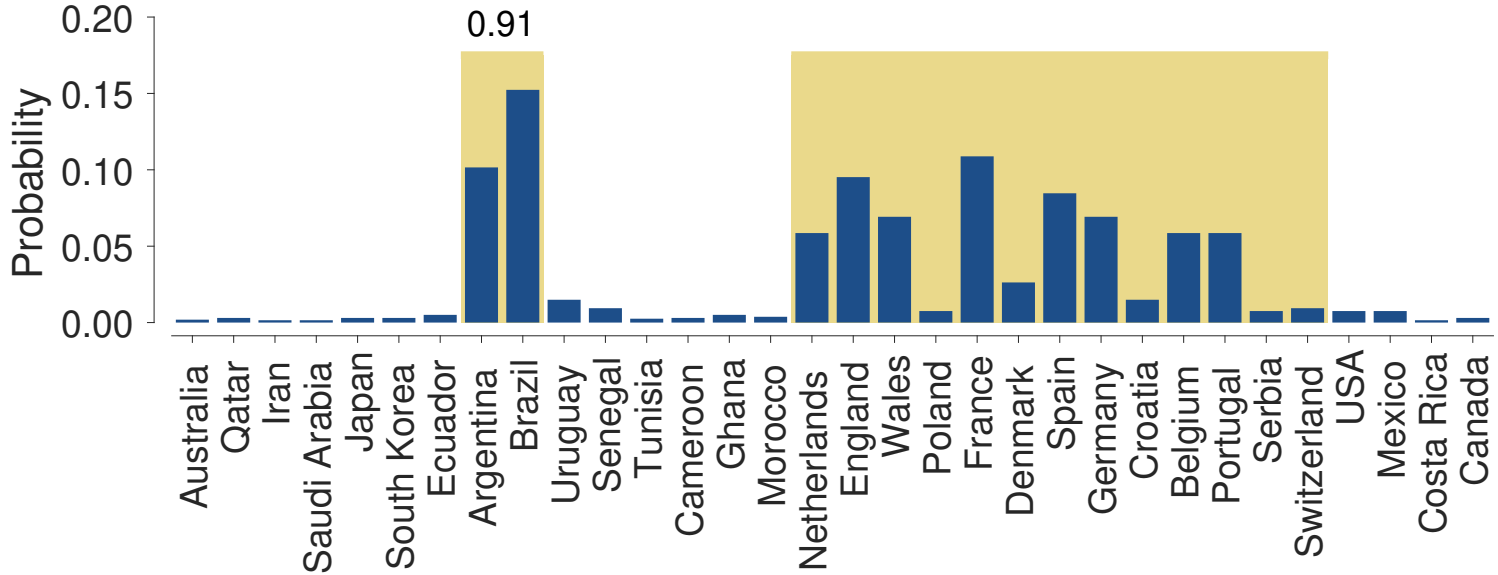
Bayesian Statistics

- Bayesian statistics are an excellent way to analyze theories, models, and data in the empirical sciences (McElreath, 2020; Vandekerckhove, Rouder, & Kruschke, 2018)
  - the key idea is that uncertainty should be fully represented in inferences and predictions, quantifying what we do and do not know
  - probability distributions provide a formal way to represent our uncertainty

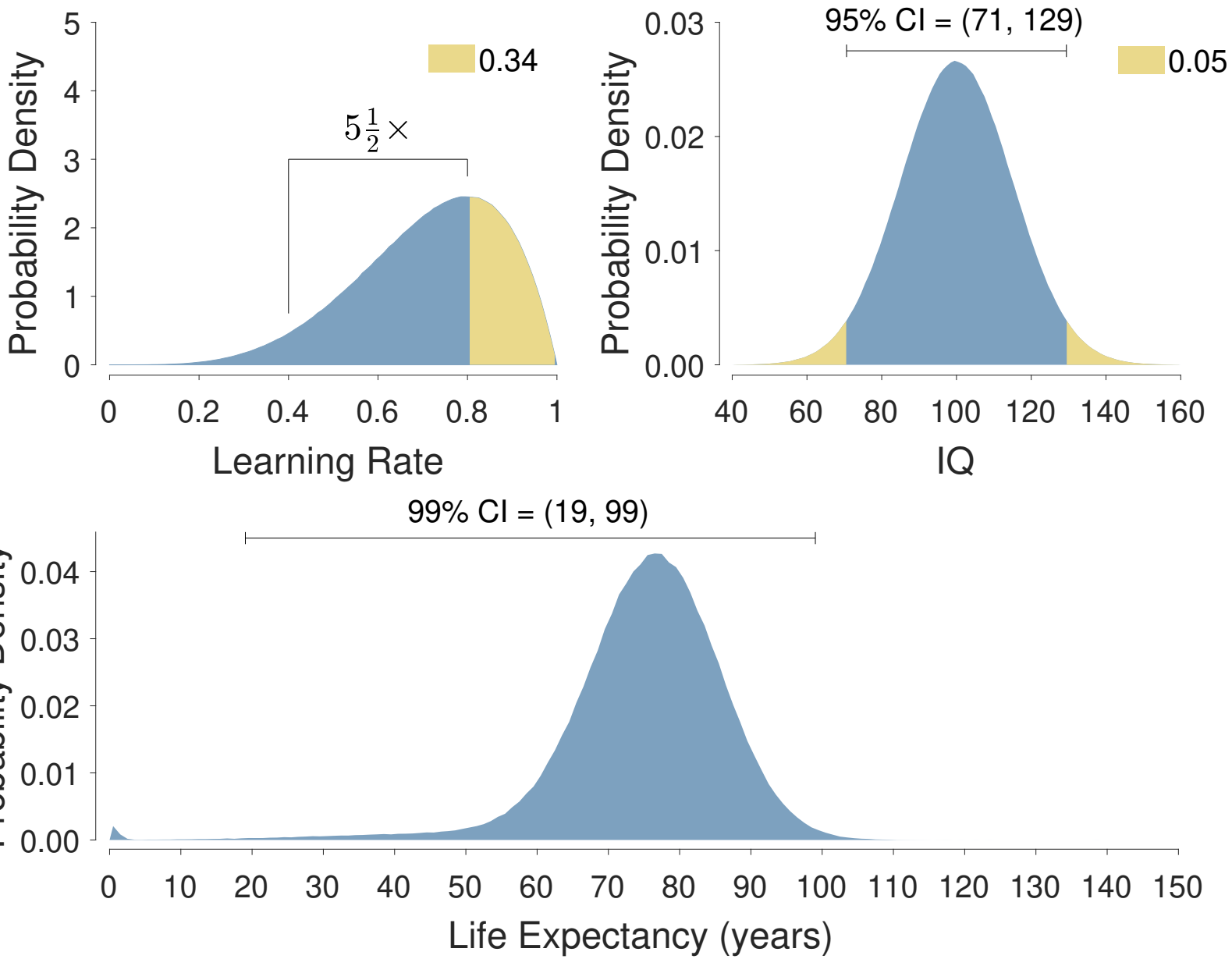
Discrete Probability Distributions





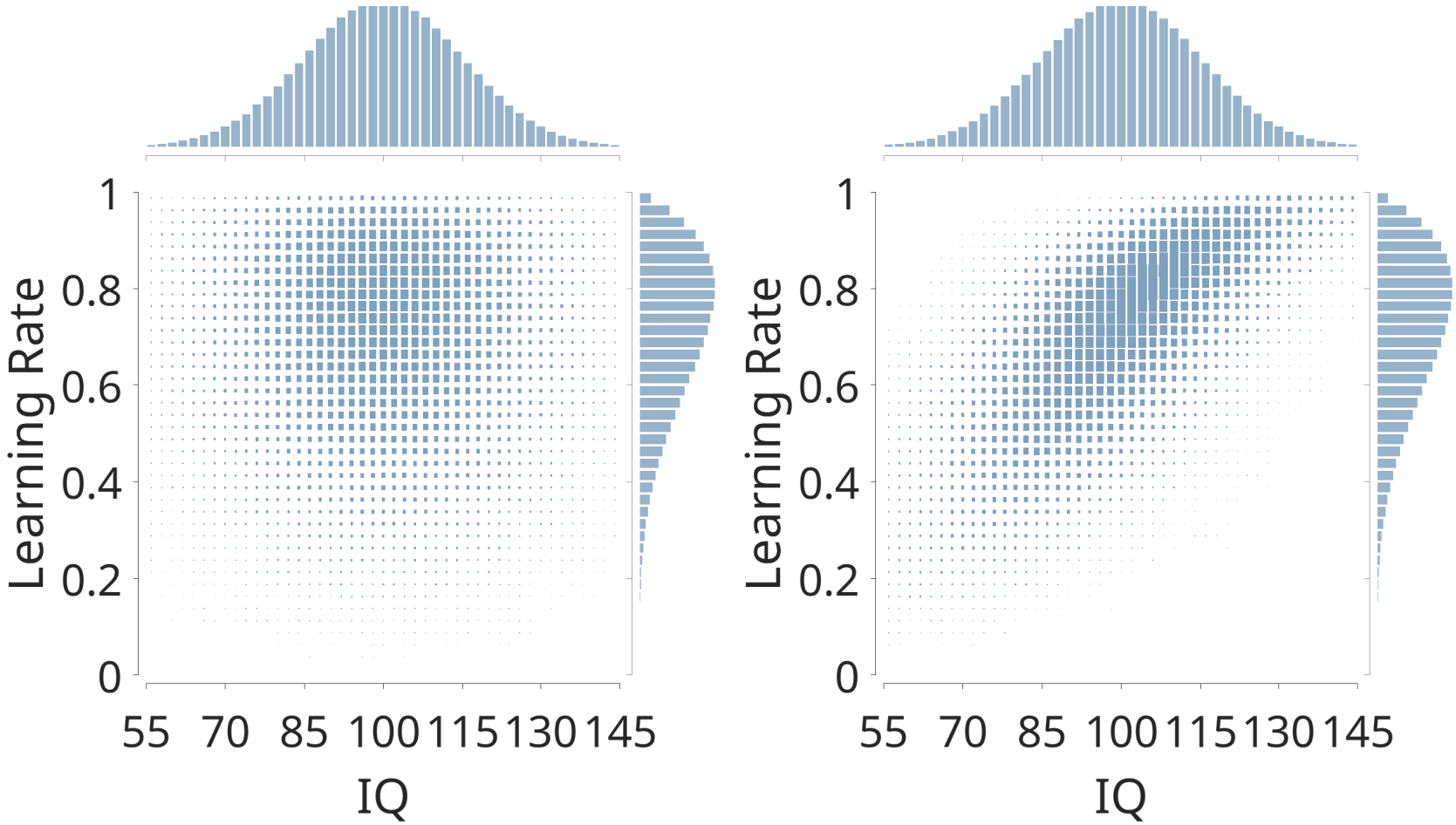


Continuous Probability Distributions



Joint and Marginal Distributions

- Joint distributions show the relative probability of every combination of two or more variables
  - can sometimes carry more information than the marginal distributions



Bayes Rule

- Bayes Rule formalizes how the evidence provided by data should be used to update probability distributions
  - prior distribution: before the data are observed
  - posterior distribution: after being updated by data

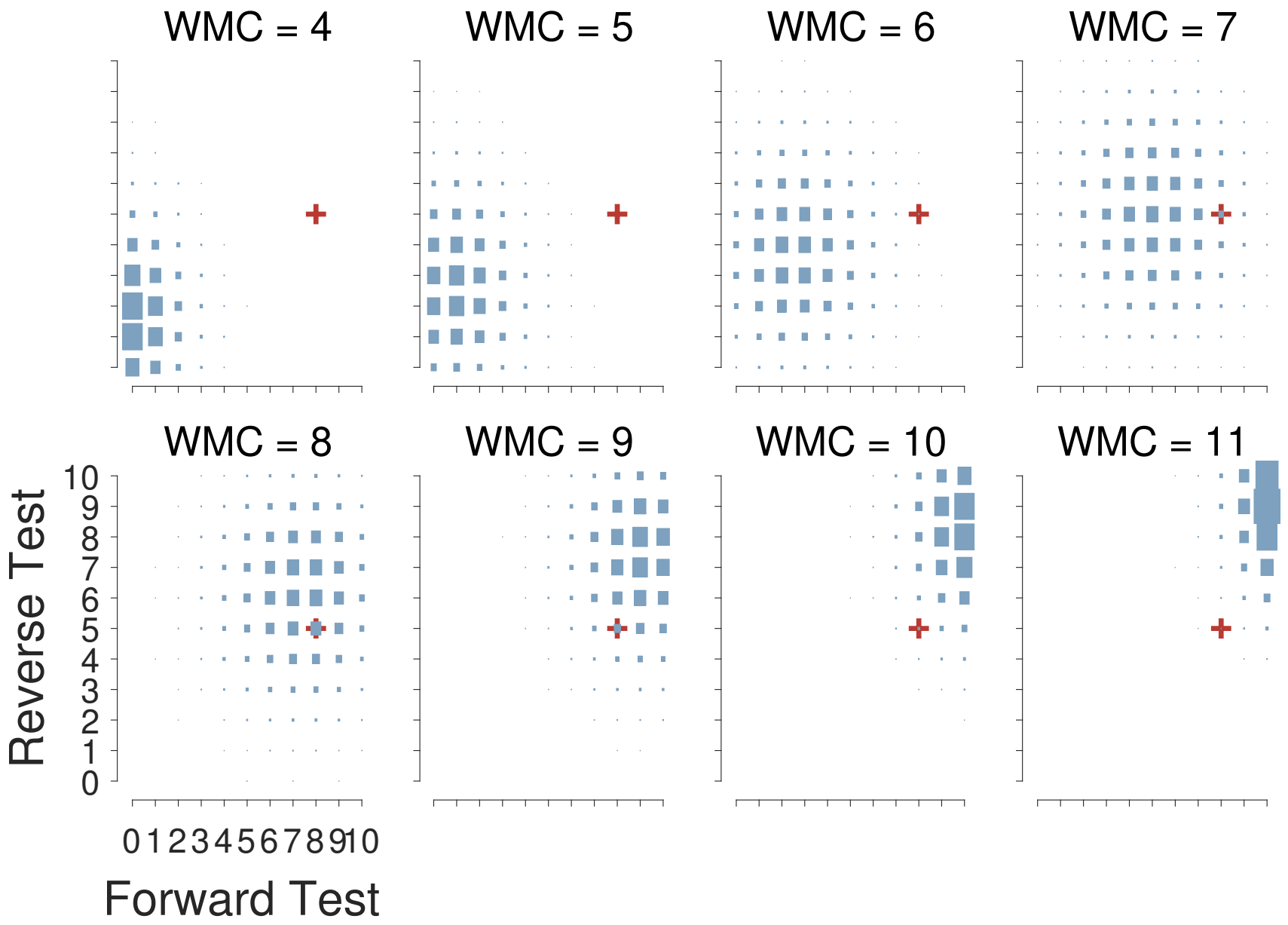
$$\underbrace{p\left(\text{parameters} \mid \text{data}\right)}_{\text{posterior distribution}} = \underbrace{p\left(\text{parameters}\right)}_{\text{prior distribution}} \times \underbrace{\frac{p\left(\text{data} \mid \text{parameters}\right)}{p\left(\text{data}\right)}}_{\substack{\text{predictive} \\ \text{updating factor}}}$$

- The predictive updating factor can be interpreted in terms of the
  - likelihood: the numerator, which quantifies the relative probability of the data arising for any given values of the parameters
  - normalizing constant: the denominator, which ensures the total probability or density is 1

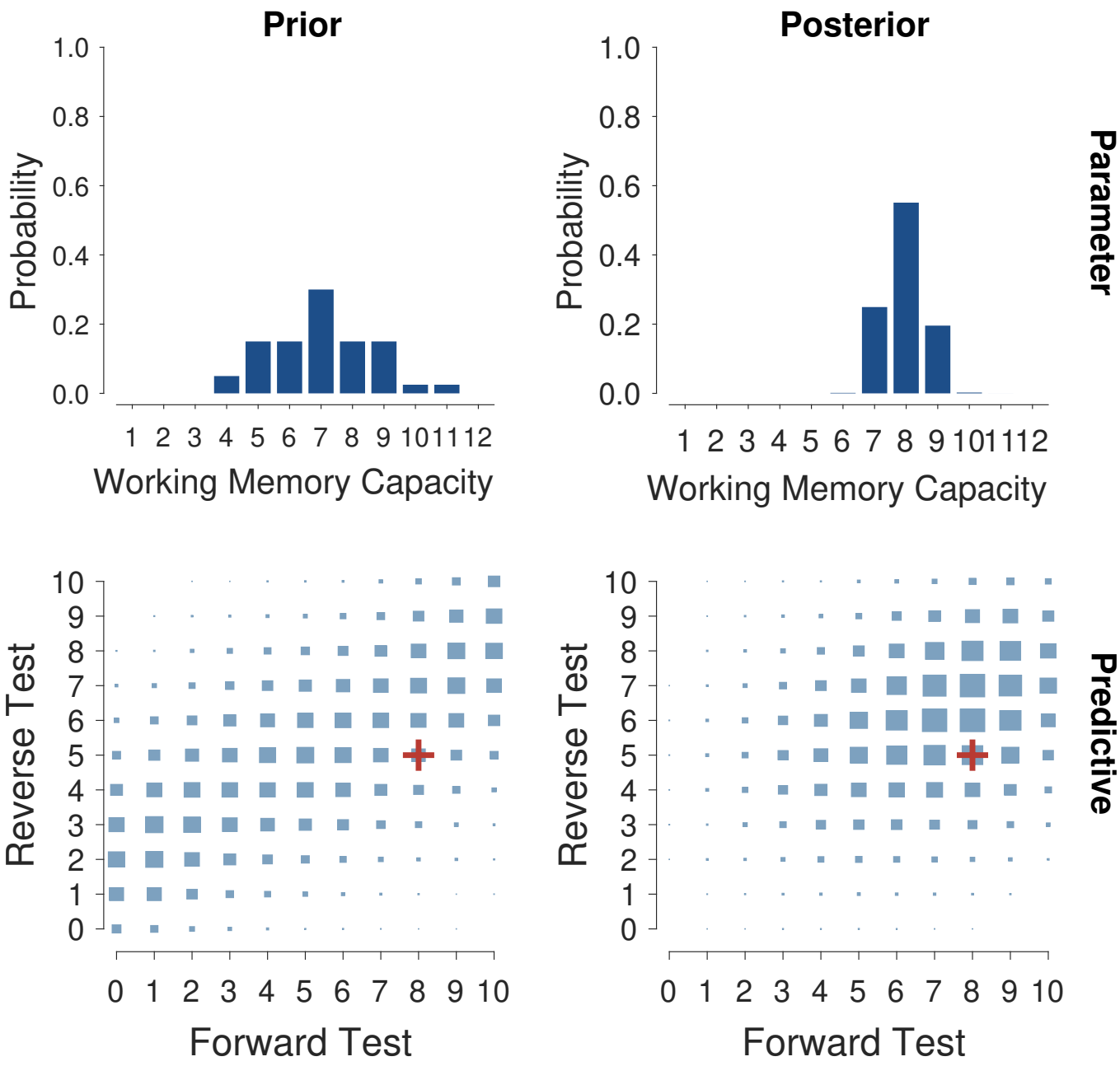
Working Memory Capacity Example

- Consider an experiment designed to measure a person's working memory capacity
  - the person taking both a forward and a reverse digit span test
  - in both tests they have to recall 10 sequences of digits and their score is the number they get correct
- The research goals are to
  - infer the working memory capacity of the person
  - make predictions about the range of their possible task performance
- In Bayesian terms, there is
  - working memory capacity parameter
  - data giving the number correct on the tasks
  - a likelihood specifying the predictions of different working memory capacities about the data

Working Memory Capacity Predictions



Working Memory Capacity Inferences



Priors, Posteriors, and Predictives

- The example highlights the logic of Bayesian inference, updating what is known about parameters based on data, and updating predictions about future data as a consequence
  - prior distribution: what is known about the possible values of a parameter before data are observed
  - prior predictive distribution: the probability of all possible data outcomes based on the prior distribution and the likelihood
  - posterior distribution: what is known about the possible values of a parameter after the data are observed
  - posterior predictive distribution: the probability of all possible data outcomes based on the posterior distribution and the likelihood

Description, Explanation, and Prediction

- The Bayesian framework allows the basic modeling goals of description, explanation, and prediction to be assessed
  - explanation: the posterior distribution of parameters, together with the likelihood, provides a psychological explanation for observed data
  - prediction: the prior predictive distributions makes predictions about data
  - description the posterior predictive distributions can be used instead to assess descriptive adequacy
- Note that despite the name the posterior predictive distribution is not a prediction for the observed data on which it is based

References i

McElreath, R. (2020). *Statistical rethinking: A bayesian course with examples in r and stan* (2nd ed.). Chapman and Hall/CRC. Retrieved from https://www.taylorfrancis.com/books/9780429642319 doi: 10.1201/9780429029608

Vandekerckhove, J., Rouder, J. N., & Kruschke, J. K. (2018). Editorial: Bayesian methods for advancing psychological science. *Psychonomic Bulletin & Review*, *25*, 1531–5320.